Course Title: **Geographic Information System**

Course No. : ICT. Ed. 473	Nature of course: Theoretical + Practical
Level: Bachelor	Credit hours: 3 (2T+1P)
Semester: Seven	Teaching hours: 64 (32T+32P)

1. Course Description

The aim of the course is to impart knowledge of the basic concepts of geographic information system (GIS) and to help the students build skills for solving problems using it. It provides the students with the basic features of the GIS such as spatial data, reading, analyzing and interpreting maps, GIS data models, finding information in raster system and vector system. It also provides knowledge about pattern analysis which includes networks, statistical surface and topological surfaces. Students are more engaged in laboratory work to realize GIS experiments rather than theoretical concept.

2. General Objectives of the Course

Following are the general objective of this course:

- To make the student knowledgeable about the geographic information system concept.
- To enable the student in implement the map analysis in practices.
- To acquaint the student in organization of geographic objects and to locate them in map.
- To explore the raster model, vector model and representing surface with these models.
- To provide the students with the skills of using GIS tool to solve the real world problems.

3. Specific Objectives and Contents

Specific Objectives	Contents
• Describe geographic information system and its scope. • Explain spatial data and how to think spatially. • Install and configure QGIS app. • Learn to use QGIS interface.	**Unit 1: Introduction to GIS** **[8]** 1.1 GIS Introduction 1.2 Scope of GIS 1.3 Think Spatially **<u>Practical Works</u>** • *Installing QGIS* • *Running QGIS for the first time* • *Introducing the QGIS user interface* • *Finding help and reporting issues*

• Categorizing the space on a map. • Understand the levels of measurement and relationship between data measurement and symbology. • Recognize, analyze, quantify patterns and make decisions. • Make use of QGIS to load raster data, vector data from files and style the layers • Create new vector layers and edit vector geometries	**Unit 2: Reading, Analyzing and Interpreting Maps [12]** 2.1 Space Categorization on a map 2.2 Levels of measurement 2.3 Relationship between symbology and data measurement 2.4 Pattern Recognition - Random, Clustered, Uniform distributional patterns 2.5 Pattern Analysis and Quantification 2.6 Result Interpretation and Decision Making **<u>Practical Works</u>** Use QGIS Application to perform following task: • *Loading vector data from files* • *Loading raster files* • *Styling raster layers* • *Styling vector layers* • *Creating new vector layers* • *Editing vector geometries* • *Editing attributes*
• Describe GIS data models • Elaborate Raster model and vector model. • Represent surface in raster and vector models. • Use QGIS tool to analyze raster data, combine raster and vector data. • Design printing maps and present map online.	**Unit 3: GIS Data Model [14]** 3.1 Raster Model and Structure 3.2 Vector Representation 3.3 Surface Representation in Raster Model 3.4 Surface Representation in Vector Model **<u>Practical Works</u>** Use QGIS Application to perform following task: • *Analyzing raster data* • *Combining raster and vector data* • *Leveraging the power of spatial databases* • *Advanced vector styling* • *Labeling* • *Designing print maps* • *Presenting your maps online*

• Define Geographic objects. • Demonstrate searching different geographic objects in GIS. • Extract, transform and load vector data and visualize GIS data. • Make use of Postgres with PostGIS and pgRouting. • Elaborate database importing and topological relationships. • Establish travel time isochron polygons.	**Unit 4: Searching for Geographic Objects [12]** 4.1 Finding Information in Raster Systems 4.2 Finding Features in Vector Systems 4.3 Searching Polygons in a GIS 4.4 Locating 2-D Map Objects 4.5 Defining the Groups for Searching **<u>Practical Works</u>** Use QGIS Application to perform following task: • *Acquiring data for geospatial applications* • *Visualizing GIS data* • *Vector data – Extract, Transform, and Load* • *Raster analysis* • *Publishing the results as a web application* • *Postgres with PostGIS and pgRouting* • *OpenStreetMap data for topology* • *Database importing and topological relationships* • *Creating the travel time isochron polygons*
• Clarify the concept of distance measurement. • Analyze different geographic patterns • Explain statistical surface, topological surface and networks. • Measure connectivity and direct traffic in roads • Make use of Road graph plugin. • Calculate the shortest paths using the Road graph plugin • Visualize pgRouting result in QGIS tool	**Unit 5: Geographic Pattern Analysis [18]** 5.1 Distance Measurement -absolute, relative, functional distance 5.2 Statistical Surfaces Characteristics, working with surface data, predicting values with interpolation 5.3 Topological Surfaces 5.4 Networks - Connectivity measurement, impedance values, one way paths, circuits, turns and intersections, directing traffic and exploiting networks **<u>Practical Works</u>** Use QGIS Application to perform following task: • *Creating a simple routing network*

• Automate multiple route computation using batch processing	• *Calculating the shortest paths using the Road graph plugin* • *Routing with one-way streets in the Road graph plugin* • *Calculating the shortest paths with the QGIS network analysis library* • *Routing point sequences* • *Automating multiple route computation using batch processing* • *Matching points to the nearest line* • *Creating a routing network for pgRouting* • *Visualizing the pgRouting results in QGIS* • *Using the pgRoutingLayer plugin for convenience* • *Getting network data from the OSM*

Note: The figures in square brackets indicate approximate teaching hours allotted to respective units.

4. General Instructional Techniques

Lecture preferably with the use of multi-media projector, demonstration, practical classes, discussion, and brain storming in all units as far as practicable.

4.1 Specific Instructional Techniques

Demonstration is an essential instructional technique for all units in this course during teaching-learning process. Specifically, demonstration with practical works will be specific instructional technique in this course. The details of suggested instructional techniques are presented below:

Units	Activities
Unit 1 to 5	• QGIS tool is used to elaborate each units concepts • Monitoring of students' work by reaching each student and providing feedback for improvement • Presentation by students followed by peers' comments and teacher's feedback • Demonstration by the teacher on practical works mentioned in each unit • Lab work individually or in pairs is assigned by the teacher to understand each unit • Assignment should be assigned to prepare lab report/project report for individual student

5. Evaluation

Evaluation of students' performance is divided into parts: Internal assessment and internal and external practical examination and theoretical examinations. The distribution of points is given below:

Internal Assessment	Internal and External Practical Exam/Viva	Semester Examination (Theoretical exam)	Total Points
40 Points	20 Points	40 Points	100 Points

***Note**: Students must pass separately in internal assessment, external practical exam and semester examination.*

5.1 Internal Evaluation (40 Marks):

Internal evaluation will be conducted by subject teacher based on following criteria:

• Class Attendance	5 Marks
• Learning activities and class performance	5 Marks
• First assignment (written assignment)	10 Marks
• Second assignment (Case Study/project work with presentation)	10 Marks
Total	40 Marks

7.1 Semester Examination (40 Marks)

Examination Division, Dean office will conduct final examination at the end of semester.	
• Objective question (Multiple choice questions 10 x 1 point)	10 Points
• Short answer questions (6 questions x 5 marks)	30 Points
Total	40 points

5.2 **Practical Exam/Viva (20 Points)**

Internal assessment (Record Book-4 points, Project work Presentation-2, Internal Practical Test-2 Points)	Semester final examination	Total
8 Points	12 Points	20 Points

5.3 Recommended Books and References materials (including relevant published articles in national and international journals)

5.4 Prescribed Textbook

DeMers, M. N. *GIS For Dummies,* For Dummies

Graser, A. et al. *QGIS Becoming a GIS Power User*-Packt Publisher

5.5 Recommended Books

Bolstad, P. & Manson, S. (2022). *GIS Fundamentals A first text on Geographic information systems (7th Ed.)*, Eider Press

Bearman, N. (2021). *GIS Research Methods (1st Ed.)*, Bloomsbury Academic

Wegmann, M. et al. (2020). *An introduction to spatial data analysis (1st Ed.)*, Pelagic